LABORATORY FOR HEALTH, OMICS AND PATHWAY EXPLORATION

Internal Training Document

STUDENT REFERENCE & INTERNAL USE NOTICE

This protocol documents standard microbial culture media selection, colony isolation, purity assessment, and preservation procedures used in HOPE Lab research and is provided to lab members for training and internal reference. External users should independently assess method suitability, verify current standards, and obtain required biosafety approvals. This document does not replace an institutionally approved SOP or method-specific risk assessment.

TITLE

Microbial Culture Media, Colony Isolation, Purification and Preservation Protocol

PRINCIPLE

Microbiological research depends on the ability to recover a target organism in a suitable culture system, separate it from other organisms, assess the purity and identity of the resulting isolate, and preserve a traceable stock for future work (Hossain, 2024). Culture media provide nutrients, salts, buffering capacity, and selective or differential components. Colony isolation separates colony-forming units (CFUs) from a mixed population. A visible colony is conventionally treated as arising from one CFU; however, a CFU may consist of a single cell or a group of attached cells (e.g., pairs, chains, or clumps). Therefore, a single colony does not necessarily demonstrate that the original inoculum contained only one cell.

Purity assessment combines observations of colony phenotype and microscopic characteristics with additional confirmation appropriate to the organism. Preservation maintains viability and relevant biological characteristics while ensuring that the identity, passage history, and storage location of each culture remain completely traceable.

Learning Outcomes

- Explain how medium selection depends on the target organism and specific research question.
- Distinguish colony isolation techniques (streak-plate) from quantitative enumeration methods (spread/pour-plate).
- Recognise the inherent limitations of colony morphology and microscopy as sole evidence of culture purity.
- Identify the essential records, quality checks, and controls required for culture authentication and cryopreservation.
- Maintain complete traceability of isolates, working stocks, master reference stocks, and passage history.

OBJECTIVE & SCOPE

Objective: To describe the standardized principles and quality requirements for culture-medium selection, microbial isolation, purity assessment, strain authentication, and preservation of microbial cultures (Hossain et al., 2020; Paul et al., 2024).

Scope: Applicable to culturable aerobic and facultative anaerobic bacteria and fungi isolated from food, environmental, clinical, and probiotic sources. Excludes obligate anaerobes and fastidious species requiring specialized atmospheres or cell systems without validated protocol modifications.

CORE MEDIA & ISOLATION CONCEPTS

- **Media Classification & Selection:** General-purpose media (TSA, NA) support non-fastidious organisms; selective media (MacConkey, MSA) suppress unwanted flora; differential media distinguish metabolic phenotypes; enriched media support fastidious growth.
- **Isolation vs. Quantitative Enumeration:** Streak-plating isolates individual CFUs for pure culture generation. Spread-plating and pour-plating quantitatively estimate viable CFUs per unit volume (CFU/mL or CFU/g).
- **Purity Assessment Limits:** Identical colony phenotypes or microscopic features do not confirm species/strain identity. Colony morphology provides preliminary evidence only; formal authentication requires molecular or biochemical confirmation.
- **Preservation & Traceability:** Cryopreservation at -80°C in 20–30% glycerol maintains cell viability. Dual-stock management (Master vs. Working stocks) minimizes genetic drift caused by repeated subculturing.

MATERIALS & EQUIPMENT

Reagents & Consumables

- Culture media (NA, TSA, MRS, MacConkey)
- Sterile diluent (0.85% Saline / PBS / MRD)
- Inoculation loops, needles & glass spreaders
- Sterile 2.0 mL cryovials & 20–30% Glycerol
- Gram stain kit & glass microscope slides
- Control reference strains (ATCC standards)

Equipment & Containment

- Calibrated Autoclave (121°C, 15 psi)
- Bacteriological Incubator (35 ± 2°C / 37°C)
- Biosafety Cabinet (Class II BSL-2)
- Brightfield Microscope (100× Oil Immersion)
- Ultra-Low Temperature Freezer (-80°C)
- Vortex Mixer & Analytical Balance

STEP-BY-STEP PROCEDURE

Step 1: Medium Preparation & Quality Control (Batch Release)

Prepare selected culture media according to manufacturer instructions. Measure and record pH prior to sterilization. Autoclave at 121°C (15 psi) for 15 minutes. Cool media to 45–50°C before pouring into sterile Petri dishes (~20–25 mL per plate). Incubate uninoculated plates at 35°C for 24 hours as a sterility check, and streak standard reference strains to verify performance prior to sample testing.

Step 2: Colony Isolation (Streak-Plate Method)

Aseptically transfer a loopful of mixed sample or broth culture onto the primary quadrant of an agar plate. Flame-sterilize the loop and allow it to cool. Draw the loop 2–3 times through the primary streak and spread into the second quadrant. Repeat the flaming and streaking process for quadrants 3 and 4 to yield well-isolated single colonies (CFUs). Incubate inverted at 35 ± 2°C for 18–24 hours.

Step 3: Quantitative Enumeration (Spread-Plate Method)

Prepare serial tenfold dilutions (10^{-1} to 10^{-6}) by transferring 1.0 mL of sample into 9.0 mL of sterile 0.85% saline diluent. Pipette 0.1 mL of chosen dilutions onto the center of pre-dried agar plates. Spread evenly across the agar surface using a sterile bent glass rod (hockey stick). Incubate plates inverted for 24 hours. Count plates containing 30–300 colonies and calculate CFU/mL = (Number of colonies × Dilution factor) / Volume plated (mL).

Step 4: Purity Assessment & Strain Authentication

Inspect isolated colonies for uniform morphological characteristics (size, shape, margin, elevation, color, opacity, texture). Pick a single discrete colony and subculture onto fresh non-selective media to obtain a pure culture. Perform Gram staining to verify cellular homogeneity (Gram reaction, morphology, cell arrangement). Conduct necessary biochemical or molecular tests for strain authentication.

Step 5: Cryopreservation & Stock Management

Inoculate a pure single colony into appropriate broth and grow to late-exponential phase. Aseptically mix 0.8 mL of fresh bacterial broth culture with 0.2 mL of sterile 80% glycerol broth in a sterile 2.0 mL cryovial (final glycerol concentration 16–20% v/v). Vortex thoroughly, label with isolate ID, passage number, and date, then freeze immediately at -80°C. Establish a Master Stock (never thawed) and Working Stock system.

Step 6: Post-Preservation Recovery & Verification

To recover frozen stocks, scrape a small portion of ice from the cryovial using a sterile loop without thawing the entire tube contents. Streak onto non-selective agar and incubate at 35°C. Verify purity and phenotypic traits against original baseline records. Document all passage histories in the HOPE Lab Strain Register.

CRITICAL FACTORS AFFECTING RESULTS

Variable	Effect of Improper Control	Required Control Standard
Medium pH & Lot	Altered growth rates and false differential reactions	Verify pH (7.2–7.4 at 25°C); log lot numbers
Sterilization Cycle	Media contamination or thermal degradation of nutrients	Autoclave at 121°C for 15 min; monitor indicators
Inoculum Density	Overcrowded colonies preventing true single colony isolation	Serial dilution prior to plating; flame loop properly
Incubation Temp/Atm	Suppression of target growth or altered colony phenotypes	Calibrated incubator (35 ± 2°C); monitor daily
Purity Check Scope	Carrying mixed/contaminated strains into downstream studies	Combine colony morphology with Gram stain verification
Glycerol Concentration	Cell death from ice crystal damage (<15%) or toxicity (>30%)	Maintain 15–20% final sterile glycerol concentration

QUALITY CONTROL & TROUBLESHOOTING

Quality Control (QC): Test reference strains (e.g., E. coli ATCC 25922, S. aureus ATCC 25923) alongside sample processing runs. Maintain a media preparation log capturing batch lot numbers, pH readings, sterilization validation, and sterility test results.

Problem	Likely Cause	Corrective Action
No or faint growth	Hot loop killed inoculum, medium mismatch, or bad incubator	Allow loop to cool; check medium suitability & temp
Confluent growth (no single colonies)	Inoculum density too high; improper flaming between streaks	Repeat streak-plating; flame loop between quadrants
>1 colony / Gram morphology	Mixed culture or contamination during handling	Subculture single colony; enforce strict aseptic technique
Loss of viability after freezing	Freezer failure, wrong glycerol conc., or repeated thawing	Check freezer temp (-80°C); use working stocks only

BIOSAFETY & ADDITIONAL CONSIDERATIONS

- **Biosafety Protocols:** Perform all manipulations involving potential pathogens inside a certified Biosafety Cabinet using BSL-2 precautions. Decontaminate work surfaces with 70% ethanol before and after procedures. Autoclave all biohazard waste at 121°C for 30 minutes prior to disposal.
- **Strain Register & Repository Maintenance:** Maintain a centralized master strain register documenting isolate ID, sample source, isolation date, identity evidence, storage freezer location, vial numbers, and passage history. Record the formal disposition when any culture stock is retired or discarded.

10. REFERENCES & RELATED DOCUMENTS

Hossain TJ, Chowdhury SI, Mozumder HA, Chowdhury MNA, Ali F, et al. (2020). Hydrolytic exoenzymes produced by bacteria isolated and identified from the gastrointestinal tract of Bombay duck. Frontiers in Microbiology, 11, 1025. doi:10.3389/fmicb.2020.01025

Lapage SP, Shelton JE, Mitchell TG, MacKenzie AR. (1970). Culture collections and the preservation of bacteria. Methods in Microbiology, 3A, 135–228.

Ali F, Silvy TN, Hossain TJ, Uddin MK, Uddin MS. (2021). Prevalence and antimicrobial resistance phenotypes of Salmonella species recovered at various stages of broiler operations in Hathazari, Bangladesh. International Journal of One Health, 7(2), 158–164. doi:10.14202/IJOH.2021.158-164